

Biochemistry lab

Experiment (6)



EXPERIMENT 6

QUANTITATION OF PROTEINS

-Spectrophotometric protein? quantitation assays are method that use UV and visible spectroscopy to rapidly determine the concentration of protein, relative to a standard.

To remember:

Spectrophotometry depend on law named beer's Law ($A = a \cdot b \cdot C$)

and you know that a and b are constant so A(absorbance) $\propto$ C (concentration) so:

$$\frac{A_1}{A_2} = \frac{C_1}{C_2}$$

Linear relationship

We use

-Spectrophotometer : a device for measuring light intensity.

-Deviation form (beer's Law At low concentration)

so We have to use it at low concentration to keep $A \propto C$

-Precipitation of protein by acidic reagent(Sulphosalicyclic acid and thichloroacetic acid)

- %T: percentage of transmission $= \frac{I_t}{I_o}$

I_o : intensity of falling light(before passing through solution)

I_t : intensity of transmitted light(after passing through solution)

$$A = - \log \frac{I_t}{I_o}$$

The quantitation of protein can be studied by different spectrophotometric methods:

1- Turbidimetry

2- The Biuret assay

First: Turbidimetry

The principle:

- First we will do as in experiment 5, we will mix the protein with acidic reagent and the protein will become insoluble precipitate → turbidity and then measure this turbidity by spectrophotometer at $\lambda_{\max} = 620 \text{ nm}$

But in experiment 2 when we use the spectrophotometer we measure the absorbance of two dyes by measuring the loss of light after the substance (color) absorbs the light when passing through the solution.

But here we don't have colored compound but we depend on particles precipitate (scattered the light)

في التجربة الثانية حددنا الامتصاص A عن طريق اسقاط ضوء على عينة فاللون داخل العينة يمتص الضوء و بالتالي تختلف الشدة بسبب اللون و لكن هنا بالبروتينات لا يوجد لون لذلك سنعتمد في قياس الامتصاص على تشتت الجزيئات المترسبة للضوء و بالتالي تختلف شدته.

SO Turbidimetry : Which is the process of measuring the loss of intensity of transmitted light.

→ Due to scattering effect of particles suspended in it

"الجزيئات المترسبة يتعمل تشتت للضوء وهذا لتقليل ال I وحساب A"

- Light is passed through a filter creating a light of known wave length

- Then passed through a Cuvette containing a solution

- A photoelectric cell collects the light which pass through the cuvettes

- A measurement is then given for the amount of absorbed light

→NOTES :

(1) The precipitation will occur by acidic reagent

(2) The concentration is directly proportional to the absorbance

(3) you must take the absorbance of the solution after 10 min because if you wait for long period of time , the protein will precipitate

إذا ترسب لن يتمكن من ترسيب الضوء
also high concentration of acidic reagent

(4) To carry out this experiment, the protein concentration range in the sample must be up to 200mg / 100ml or 200mg / 1 dl (e.g. urine sample) , if the concentration > 200mg / 100ml the protein will precipitate even with low concentration of TCA or SSA

(5) $\text{mg} / 100 \text{ ml} = \text{mg} / 1 \text{ dl}$

→STEPS:

There will be 5 test tubes :

-First one is **blank** → is the reference contain only solvent and material except the substance to be measured.

-The last one for the Unknown test tube (0.5 ml)

- The 3 test tubes will be standard concentration
60,120,180 mg / dl → (0.5 ml per each)

1- Mix 0.5 ml of the standard with 2 ml of the 3% Trichloroacetic acid or 3% Sulphosalicylic acid, mix and read absorbance after 10 min at $\lambda_{\max}$ = 620 nm

Tube	Protein volume "V1"	H ₂ O	TCA OR SSA	Absorbance	C ₁
1	0	0.5 ML	2 ML	By using spectrophotometer	0
2	0.5 ML	0	2 ML		60 mg/dl
3	0.5 ML	0	2 ML		120 mg/dl
4	0.5 ML	0	2 ML		180 mg/dl
5(unknown)	0.5 ML	0	2 ML		?

يمكن استخدام قانون التخفيف في هذه التجربة كما في تجربة 2

$$V_1 C_1 = V_2 C_2$$

→ for example tube 2

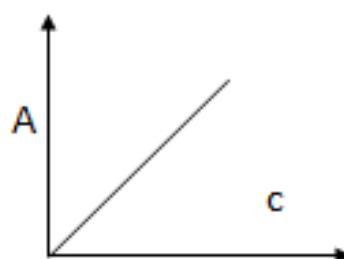
$$0.5 \times 60 = 2.5 \times C_2$$

$$C_2 = 12 \text{ mg/dl}$$

$$\text{tube 3 } 24 \text{ mg/dl}$$

$$\text{tube 4 } 36 \text{ mg/dl}$$

2- Establish the calibration curve (standard curve) – (A vs. C)
Then apply the absorbance and find the concentration of the unknown



3- Or you can calculate it by:

$$\frac{A_1}{A_2} = \frac{C_1}{C_2}$$

Second: The Biuret assay (quantitative test)

- A Method that can be used to determine the amount of soluble protein in solution.
- وهي طريقة تستخدم للتراكيز الاعلى من 200 mg/dl
- Biuret reagent (copper sulfate in a strong base with sodium potassium tartarate.)
- The Biuret Reagent reacts with peptide bonds and change color → blue-violet complex.
- * Here the spectrophotometer is used to measure the intensity of the color produced → the more protein (peptide bonds) the darker the color.

Biuret test: Alkaline copper sulfate reacts with compounds containing two or more peptide bonds (any/all proteins) to give a violet colored complex (blue-violet).

- All proteins give positive results with biuret test.
- As the # of peptide bonds increase, the depth of the color increase.

Notes:

1. We can have positive results in sample that have protein concentration range up to 800 mg/dL.
2. The intensity of the violet color is measured by spectrophotometer at $\lambda_{\text{max}} = 540\text{nm}$.
3. Tartarate salt that are found in Biuret reagent used to prevent the formation of Cu^{+2} ions as $\text{Cu}(\text{OH})_2$ (copper sulfate) ← Important

Important Question: In the quantitation of protein, what are the components of biuret reagent?

- Cu^{+2} + strong base (NaOH)
- Cu^{+2} + sodium potassium tartarate
- Cu^{+2} + strong base (NaOH) + sodium potassium tartarate
- Non of the above

Steps:-

- 1 – prepare a blank solution by mixing 3 ml of biuret reagent to 2 ml of water .
- 2- Add 3ml of biuret reagent to 2ml of each of the standards and unknown protein solution also and mix .
- 3- warm at 37 C for 10 min → cool them → record the absorbance at $\lambda_{\text{max}} = 540\text{ nm}$.

Tube	Protein volume "V1"	H ₂ O	biuret	Absorbance	C ₁
1	0	2 ML	3 ML	By using spectro-photometer	0
2	2 ML	0	3 ML		250 mg/dl
3	2 ML	0	3 ML		500 mg/dl
4	2 ML	0	3 ML		750 mg/dl
5 (unknown)	2 ML	0	3 ML		?

يمكن استخدام قانون التخفيف في هذه التجربة كما في تجربة 2:

$$V_1 C_1 = V_2 C_2$$

→ for example tube 2

$$2 \times 250 = 5 \times C_2$$

$$C_2 = 100 \text{ mg/dl}$$

tube 3 200 mg/dl

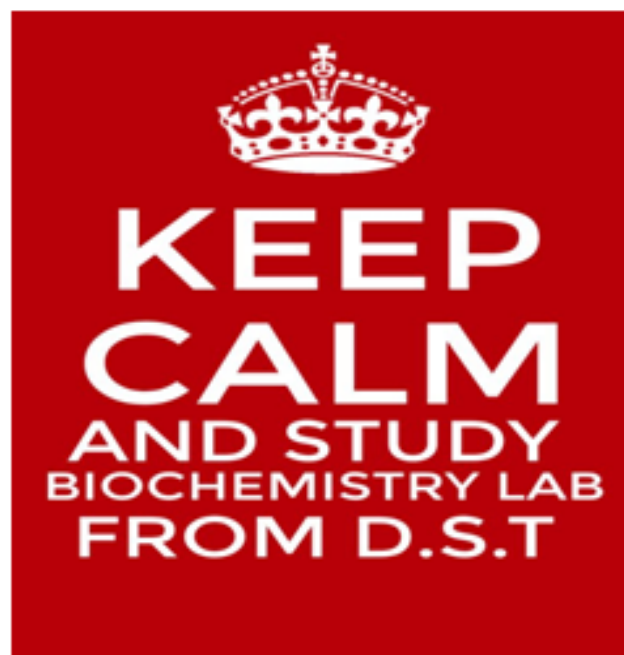
tube 4 300 mg/dl

4- prepare a graph of A against albumin concentration (standard) then find the absorbance of the unknown from the graph or by equation

$$\frac{A_1}{A_2} = \frac{C_1}{C_2}, \text{ Calculate it.}$$

DONE BY:

Hussein Alsaleh, Hisham Darwish, Dina Nasser Eddine, Lynne Abdullah, Abed Al Hadi Raed, Omar Al-Ali, Danah Alghanim and Ayham Haddad.



Questions

- 1) **The intensity of the color product in determination total protein by biuret method is proportional to the :**
 - a) Number of amino acids
 - b) Number of ionizable group in the proteins
 - c) Number of peptide bonds
 - d) None of the above
- 2) **A,B,C are three proteins with different sizes when they react with biuret reagent they give a close intense purple color suggest a method determine which is the largest protein :**
 - a) Measure the absorbance
 - b) Use a higher concentration of biuret reagent and see which one will form the largest precipitate
 - c) Make ninhydrin test
 - d) A & B
- 3) **Biuret solution contains :**
 - a) Cu^{2+}
 - b) NaOH
 - c) Sodium Potassium tartarate
 - d) All of above are correct
- 4) **All of the following statements about turbidimetry are true except :**
 - a) The precipitation will occur by acidic reagent
 - b) The process of measuring the loss of intensity of transmitted light due to scattering affect of particles suspended in it
 - c) The protein concentration must be bigger than 200 mg/dl
 - d) We measure this turbidity by spectrophotometer at $\lambda_{\text{max}}=650\text{nm}$
- 5) **tartarate salts that are found in biuret reagent used to :**
 - a) produce the blue-violent complex (causing the color change)
 - b) raise the pH of the solution
 - c) to prevent the formation of Cu^{2+} ions as $\text{Cu}(\text{OH})_2$
 - d) none of the above